Apply to be a Science Corps Fellow - Entry #3368

Name

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Gender

Female

Residence: City (State), Country

Cleveland, Ohio, USA

Nationality

Indian

When did you or do you expect to complete your PhD?

December 2025

Please tell us briefly about your university education (institution, city, country, year)

I am a PhD candidate in Physics at Case Western Reserve University, USA where I research detector instrumentation and signal analysis for rare-event searches. Before this, I earned an MSc in Physics from the Indian Institute of Technology Guwahati, India, where I worked with optical tweezers and image reconstruction. I completed my BSc in Physics at the University of Delhi, India. My academic journey from India to the U.S. has trained me to work across diverse scientific settings and cultures, and to explain complex concepts at different levels of learning. Alongside research, I have consistently sought teaching and mentorship roles that allow me to give back to society which vary in variety, from designing undergraduate physics labs, to guiding first-year students, and mentoring women in STEM. These experiences have strengthened both my scientific expertise and my ability to make science accessible to broader communities.

Why would you like to be a Science Corps Fellow?

During my PhD in experimental physics, I refined detectors, analyzed waveforms, and simulated how radiation interacts with matter. While fulfilling, some of my most meaningful moments came when I stepped outside the lab to mentor students or guiding them through scientific ideas.These experiences taught me that the value of science lies as much in discovery as in how widely it is shared. The Science Corps Fellowship reflects this conviction by combining scientific expertise with teaching and service, allowing me to contribute in a way that leaves a lasting impact.

My teaching journey began as a Physics Teaching Assistant at the University at Buffalo. It continued at Case Western Reserve University, where I supported 200+ undergraduates across five courses. I designed and delivered labs, graded assessments, and helped students master physics concepts. When the pandemic disrupted classrooms, I adapted quickly, using online tools and live demonstrations to sustain learning. I have mentored minority students through a department program, supported women in STEM, and advised international students beginning graduate school. These roles strengthened my ability to translate abstract concepts into approachable lessons and create supportive learning environments.

I have also sought opportunities to teach in resource-limited settings. As a master's student in India, I volunteered to teach the children of campus staff in a nearby village. With few materials available, we relied on simple demonstrations and everyday examples to make learning engaging. More recently, I mentored a high school student whose parents had not attended college, introducing her to university life and pathways in physics. Small interventions, like setting up a telescope so students could see Jupiter's moons, remind me how access to science can spark lifelong curiosity, just as it did in me.

I am drawn to Science Corps because of its mission to bring this kind of access to communities rich in talent but underserved in resources. Growing up in India, I experienced firsthand the uneven distribution of quality STEM education outside major cities. The CVIF's Dynamic Learning Program resonates with me because of its emphasis on creativity, hands-on learning, and sustainability. I am excited by the idea of developing low-cost experiments, interactive demonstrations, and lesson plans that teachers can continue to use long after the fellowship ends.

This fellowship is also an opportunity for growth. Teaching abroad will challenge me to refine my communication skills and develop a global perspective, essential for my development and serving students well. Science Corps offers the unique chance to contribute my skills to a community that welcomes them, and to carry forward the lessons of cultural exchange, teaching, and mentorship into every stage of my journey.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Outside of research, I've always sought ways to mentor, teach, and share science. Those experiences within and outside of structured programs have shaped how I think about education and prepared me for a fellowship like Science Corps.

As a master's student in India, I volunteered with classmates to teach the children of campus staff in a nearby village. We didn't have many resources, so we made do with what we had: showing refraction with a glass of water, or building simple circuits from spare materials. It was eye-opening to see how much curiosity those small demonstrations could spark. That experience convinced me that effective teaching isn't about expensive labs but creativity, patience, and connecting with students.

During my PhD, I've continued mentoring in different ways. I guided a high school student, the first in her family to consider college, and introduced her to university life. At Case Western Reserve University, I mentored underrepresented students in physics through the PURMS program. I worked with the Mather Mentorship Program for Women in STEM. Each student came with different needs—sometimes academic, sometimes personal—and I learned to adapt my approach so they left feeling more capable and supported.

As President of the Local Inspirational Figures Talk (LIFT) series, I organized events highlighting diverse STEM role models. I worked on the International Student Advisory Council to improve support for students adjusting to a new country. Through the National Mentoring Committee, I connected with early-career scientists nationwide. These roles taught me how to listen, collaborate, and create inclusive spaces—all essential qualities for teaching abroad.

Some of my favorite moments have come outside of classrooms. I've helped set up telescopes for public sky-watching nights and watched students' amazement as they saw Jupiter's moons for the first time. I've judged undergraduate poster sessions and encouraged students to explain their work clearly and with pride. I also know that, for many students at CVIF, I may be their first sustained interaction with a scientist. I see that as both a responsibility and a privilege: to make science exciting, to model how scientists think, and to leave behind confidence and resources that last long after my fellowship ends.

Where do you imagine yourself professionally in the future? Check all that apply.

Doing science in industry
Administrative role in industry or academia
In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the near term, I see myself finishing my PhD and moving into a role where I can put my physics training to practical use. I'm especially drawn to medical physics and imaging, where the skills I use to study rare particles—building detectors, analyzing signals, solving technical problems—can directly improve people's lives. The idea that my work could help diagnose or treat patients motivates me in a way that feels exciting and meaningful. I know I'll keep teaching and mentoring close to me. Assisting students to grow has been one of the most rewarding parts of graduate school, and I don't want that to end when I leave academia.

Two years from now, I imagine myself in a technical role, maybe at a national lab or in the medical device industry, but also still carving out time for outreach. I expect that what I would learn as a Science Corps Fellow—how to adapt science for different audiences and make lessons work even when resources are limited—will keep shaping how I mentor students and share science in whatever setting I'm in.

In ten years, I plan to be in a leadership role that blends science with people. I don't just want to push projects forward; I want to help shape inclusive teams and open doors for students who might not otherwise see themselves in STEM. Whether that means running a research group, coordinating extensive collaborations, or even starting programs that expand access to science education, I want to be in a position where I can amplify opportunities for others.

What ties these goals together is a commitment to equity in education. My path—from studying physics in India to pursuing a PhD in the U.S.—was shaped by mentors who encouraged me to reach further than I thought possible. Ten years from now, I want to be able to offer that same encouragement on a larger scale. Science Corps isn't just a six-month fellowship; it's a chance to grow into the kind of scientist and mentor I want to become, while helping students discover the joy of science in their own lives.

Please upload your CV (.pdf)

Aneesha_Avasthi_CV_Science_Corps.pdf

How did you find out about Science Corps?

Referred by a friend/colleague/supervisor in academia

Science Corps